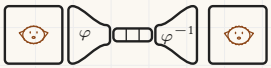
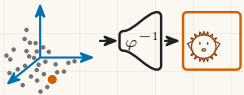
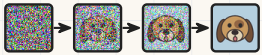
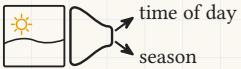
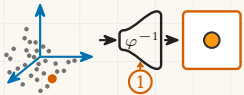
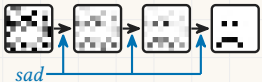
An atlas of the weekend's models

	Reads a picture	Makes a picture	
	into a few numbers makes nothing new	in one jump fast, a short code to edit	in many small steps sharper, but slow
Trained on **pictures only** no labels needed	**Autoencoder** *Friday 08:15*  3 numbers per fish	**VAE** *Friday 08:15 and 09:00*  + one cloud to pick from	**Diffusion** *Friday 11:00*  from noise, step by step
Trained on **pictures and their labels** a label decides what comes out	**CNN** *Friday 08:15*  one label per photo	**Conditional VAE** *Friday 10:00*  + a label: apple or orange	**Conditional diffusion** *Friday 11:00 and 14:00*  + a prompt at every step

Which one, when?

	Use it when	Don't use it when	Pros	Cons
Autoencoder	you want a few numbers per picture	you want new pictures	no labels needed; fast	about half its codes give smudges
CNN	you want a number or a class from a picture	you have no labels	fast; easy to test (2.7 min miss)	needs labels; tiny changes fool it
VAE	you want new pictures, fast	you need sharp detail or control	any point of the cloud is a picture	softer pictures; chance decides
Conditional VAE	you change one labelled trait and keep the rest	nobody labelled that trait	you pick what comes out; fast	needs labels; 83 of 5000 flips failed
Diffusion	quality matters more than speed	you need speed, or a code to edit	the sharpest, most varied pictures	slow: 20 passes in our notebook
Conditional diffusion	you want pictures from words	the request must be exact	you ask in words	slow; captions needed; convincing fakes